

DBS Lab 12 Tasks

1)

```
> use Employeesdb
< switched to db Employeesdb
> db.employees.insertOne({
  "name": "Ahmed Ali",
  "age": 28,
  "department": "Computer Science",
  "salary": 110000,
  "skills": ["C", "Python", "SQL"],
  "address": {
    "street": "Street 6",
    "city": "Karachi",
    "country": "Pakistan",
    "postalCode": 75600
  }
})
< {
  acknowledged: true,
  insertedId: ObjectId('69f6e9192327c979053f8e84')
}
```

2)

```

> db.employees.insertMany([
  { "name": "Usman Roshan", "department": "HR", "salary": 80000 },
  { "name": "Zainab Aziz", "department": "Marketing", "salary": 95000 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f6e9ba2327c979053f8e85'),
    '1': ObjectId('69f6e9ba2327c979053f8e86')
  }
}

```

3)

```

> db.employees.insertOne({
  "name": "Bilal Boi",
  "projects": [
    { "title": "Todo List", "status": "In Progress" },
    { "title": "Gallery Site", "status": "Completed" }
  ]
})
< {
  acknowledged: true,
  insertedId: ObjectId('69f6ea332327c979053f8e87')
}
Atlas atlas-xkko7q-shard-0 [primary] Employeesdb>

```

4)

```
> db.employees.find({})
< {
  _id: 101,
  name: 'Ali Khan',
  age: 30,
  email: 'ali.khan@example.com',
  department: 'Software Engineering',
  salary: 120000,
  isActive: true,
  skills: [
    'Java',
    'Python',
    'MongoDB',
    'Docker'
  ],
  address: {
    street: 'Street 10',
    city: 'Karachi',
    country: 'Pakistan',
    postalCode: 75200
  },
  projects: [
    {
      projectId: 'P01',
      title: 'E-commerce Website',
      status: 'Completed',
      budget: 500000
    },
  ],
}
```

```
{
  projectId: 'P02',
  title: 'AI Chatbot',
  status: 'In Progress',
  budget: 300000
}
],
joiningDate: '2022-01-15'
}
{
  _id: ObjectId('69f6e9192327c979053f8e84'),
  name: 'Ahmed Ali',
  age: 28,
  department: 'Computer Science',
  salary: 110000,
  skills: [
    'C',
    'Python',
    'SQL'
  ],
  address: {
    street: 'Street 6',
    city: 'Karachi',
    country: 'Pakistan',
    postalCode: 75600
  }
}
{
```

```

}
{
  _id: ObjectId('69f6e9ba2327c979053f8e85'),
  name: 'Usman Roshan',
  department: 'HR',
  salary: 80000
}
{
  _id: ObjectId('69f6e9ba2327c979053f8e86'),
  name: 'Zainab Aziz',
  department: 'Marketing',
  salary: 95000
}
{
  _id: ObjectId('69f6ea332327c979053f8e87'),
  name: 'Bilal Boi',
  projects: [
    {
      title: 'Todo List',
      status: 'In Progress'
    },
    {
      title: 'Gallery Site',
      status: 'Completed'
    }
  ]
}

```

Atlas atlas-xkko7q-shard-0 [primary] Employeesdb> |

5)

```
> db.employees.find({ "salary": { "$gt": 100000 } })
< {
  _id: 101,
  name: 'Ali Khan',
  age: 30,
  email: 'ali.khan@example.com',
  department: 'Software Engineering',
  salary: 120000,
  isActive: true,
  skills: [
    'Java',
    'Python',
    'MongoDB',
    'Docker'
  ],
  address: {
    street: 'Street 10',
    city: 'Karachi',
    country: 'Pakistan',
    postalCode: 75200
  },
  projects: [
    {
      projectId: 'P01',
      title: 'E-commerce Website',
      status: 'Completed',
      budget: 500000
    },
  ],
}
```

```

    {
      projectId: 'P02',
      title: 'AI Chatbot',
      status: 'In Progress',
      budget: 300000
    }
  ],
  joiningDate: '2022-01-15'
}
{
  _id: ObjectId('69f6e9192327c979053f8e84'),
  name: 'Ahmed Ali',
  age: 28,
  department: 'Computer Science',
  salary: 110000,
  skills: [
    'C',
    'Python',
    'SQL'
  ],
  address: {
    street: 'Street 6',
    city: 'Karachi',
    country: 'Pakistan',
    postalCode: 75600
  }
}

```

Atlas atlas-xkko7q-shard-0 [primary] Employeesdb>

6)

```
> db.employees.find({ "address.city": "Karachi" })
< {
  _id: 101,
  name: 'Ali Khan',
  age: 30,
  email: 'ali.khan@example.com',
  department: 'Software Engineering',
  salary: 120000,
  isActive: true,
  skills: [
    'Java',
    'Python',
    'MongoDB',
    'Docker'
  ],
  address: {
    street: 'Street 10',
    city: 'Karachi',
    country: 'Pakistan',
    postalCode: 75200
  },
  projects: [
    {
      projectId: 'P01',
      title: 'E-commerce Website',
      status: 'Completed',
      budget: 500000
    },
  ],
}
```



```

    {
      projectId: 'P02',
      title: 'AI Chatbot',
      status: 'In Progress',
      budget: 300000
    }
  ],
  joiningDate: '2022-01-15'
}
{
  _id: ObjectId('69f6e9192327c979053f8e84'),
  name: 'Ahmed Ali',
  age: 28,
  department: 'Computer Science',
  salary: 110000,
  skills: [
    'C',
    'Python',
    'SQL'
  ],
  address: {
    street: 'Street 6',
    city: 'Karachi',
    country: 'Pakistan',
    postalCode: 75600
  }
}

```

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7)

```
> db.employees.find({ "skills": "Python" })
< {
  _id: 101,
  name: 'Ali Khan',
  age: 30,
  email: 'ali.khan@example.com',
  department: 'Software Engineering',
  salary: 120000,
  isActive: true,
  skills: [
    'Java',
    'Python',
    'MongoDB',
    'Docker'
  ],
  address: {
    street: 'Street 10',
    city: 'Karachi',
    country: 'Pakistan',
    postalCode: 75200
  },
  projects: [
    {
      projectId: 'P01',
      title: 'E-commerce Website',
      status: 'Completed',
      budget: 500000
    },
  ],
}
```

```

    {
      projectId: 'P02',
      title: 'AI Chatbot',
      status: 'In Progress',
      budget: 300000
    }
  ],
  joiningDate: '2022-01-15'
}
{
  _id: ObjectId('69f6e9192327c979053f8e84'),
  name: 'Ahmed Ali',
  age: 28,
  department: 'Computer Science',
  salary: 110000,
  skills: [
    'C',
    'Python',
    'SQL'
  ],
  address: {
    street: 'Street 6',
    city: 'Karachi',
    country: 'Pakistan',
    postalCode: 75600
  }
}

```

Atlas atlas-xkko7q-shard-0 [primary] Employeesdb ›

8)

```
> db.employees.find({}, { "name": 1, "department": 1, "_id": 0 })
< {
  name: 'Ali Khan',
  department: 'Software Engineering'
}
{
  name: 'Ahmed Ali',
  department: 'Computer Science'
}
{
  name: 'Usman Roshan',
  department: 'HR'
}
{
  name: 'Zainab Aziz',
  department: 'Marketing'
}
{
  name: 'Bilal Boi'
}
```

9)

```
> db.employees.find({ "projects.status": "In Progress" })
< {
  _id: 101,
  name: 'Ali Khan',
  age: 30,
  email: 'ali.khan@example.com',
  department: 'Software Engineering',
  salary: 120000,
  isActive: true,
  skills: [
    'Java',
    'Python',
    'MongoDB',
    'Docker'
  ],
  address: {
    street: 'Street 10',
    city: 'Karachi',
    country: 'Pakistan',
    postalCode: 75200
  },
  projects: [
    {
      projectId: 'P01',
      title: 'E-commerce Website',
      status: 'Completed',
      budget: 500000
    },
  ],
}
```

```

    {
      projectId: 'P02',
      title: 'AI Chatbot',
      status: 'In Progress',
      budget: 300000
    }
  ],
  joiningDate: '2022-01-15'
}
{
  _id: ObjectId('69f6ea332327c979053f8e87'),
  name: 'Bilal Boi',
  projects: [
    {
      title: 'Todo List',
      status: 'In Progress'
    },
    {
      title: 'Gallery Site',
      status: 'Completed'
    }
  ]
}

```

Atlas atlas-xkko7q-shard-0 [primary] Employeesdb ➤

10)

```
> db.employees.updateOne(
  { "name": "Ali Khan" },
  { "$set": { "salary": 130000 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

11)

```
> db.employees.updateOne(
  { "name": "Ali Khan" },
  { "$addToSet": { "skills": "AWS" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

12)

```
> db.employees.updateOne(  
  { "name": "Ali Khan" },  
  { "$set": { "address.city": "Lahore" } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

13)

```
> db.employees.updateMany(  
  { "department": "Software Engineering" },  
  { "$set": { "isActive": false } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

14)


```
> db.employees.deleteOne({"name" : "Ali Khan"})
< {
  acknowledged: true,
  deletedCount: 1
}
```

15)

```
> db.employees.deleteMany({"salary" : {"$lt" : 50000}})
< {
  acknowledged: true,
  deletedCount: 0
}
```